```sql
CREATE DATABASE Library_Management_System;
use Library_Management_System;

Create TABLE Student(
Student_ID INT PRIMARY KEY IDENTITY(1,1),
Student_NAME VARCHAR(100),
Phone VARCHAR(11),
Email VARCHAR(100),
);
Create TABLE BOOK(
Book_Name VARCHAR (100),
Book_ID INT PRIMARY KEY IDENTITY (1,1),
);
Create TABLE Administration(
Admin_ID INT PRIMARY KEY IDENTITY (1,1),
Admin_Name VARCHAR (100),
Phone VARCHAR (11),
Email VARCHAR(100),
);
Create TABLE Reservation(
Reservation_ID INT PRIMARY KEY IDENTITY (1,1),
Student_ID INT NOT NULL,
Book_ID INT NOT NULL,
Reservation_Date DATE  NOT NULL,
Status VARCHAR(30) DEFAULT 'pending',
 FOREIGN KEY (Student_ID) REFERENCES Student(Student_ID),
    FOREIGN KEY (Book_ID) REFERENCES Book(Book_ID)
);
CREATE TABLE Fine (
    Fine_ID INT PRIMARY KEY IDENTITY(1,1),
    Student_ID INT NOT NULL,
    Book_ID INT NOT NULL,
    Amount DECIMAL(10,2) NOT NULL,
    Fine_Date DATE NOT NULL,
    Status VARCHAR(30) DEFAULT 'Unpaid',

    FOREIGN KEY (Student_ID) REFERENCES Student(Student_ID),
    FOREIGN KEY (Book_ID) REFERENCES Book(Book_ID)
);

INSERT INTO Student
(Student_Name, Email, Phone )
VALUES
('Umair Iftikhar', 'umair@gmail.com', '03001234567' ),
('Ali Khan', 'ali@gmail.com', '03111234567' ),
('Ahmed Raza', 'ahmed@gmail.com', '03221234567');

INSERT INTO Book
(Book_Name)
```

```sql
VALUES
('Computer Learning'),
('Football Technique'),
('SQL Query');

INSERT INTO Administration
(Admin_Name, Email, Phone)
VALUES
('Admin 1', 'admin1@library.com', '03001111111'),
('Admin 2', 'admin2@library.com', '03002222222');

INSERT INTO Reservation
(Student_ID, Book_ID, Reservation_Date, Status)
VALUES
(1, 2, GETDATE(), 'Pending');

INSERT INTO Fine
(Student_ID, Book_ID, Amount, Fine_Date, Status)
VALUES
(1, 2, 500.00, GETDATE(), 'Unpaid');
SELECT
    R.Reservation_ID,
    S.Student_Name,
    B.Book_Name,
    R.Reservation_Date,
    R.Status
FROM Reservation R
JOIN Student S ON R.Student_ID = S.Student_ID
JOIN Book B ON R.Book_ID = B.Book_ID;
SELECT
    F.Fine_ID,
    S.Student_Name,
    B.Book_Name,
    F.Amount,
    F.Fine_Date,
    F.Status
FROM Fine F
JOIN Student S ON F.Student_ID = S.Student_ID
JOIN Book B ON F.Book_ID = B.Book_ID;

ALTER TABLE Book
ADD Author_Name VARCHAR(100);
SELECT * FROM Book;
TRUNCATE TABLE Fine;
select * From Fine;


DROP TABLE Fine;
select * From Fine;
```

```sql

select * from Student;

UPDATE Student
SET Phone = '03123456789'
WHERE Student_ID = 1;
select * from Student;

DELETE FROM Student
WHERE Student_ID = 3;
select * from Student;

CREATE USER LibraryUser WITHOUT LOGIN;


GRANT SELECT ON dbo.Student TO LibraryUser;

REVOKE SELECT ON dbo.Student FROM LibraryUser;

BEGIN TRANSACTION;
UPDATE Fine
SET Status = 'Paid'
WHERE Fine_ID = 1;
COMMIT TRANSACTION;
select * From Fine;

BEGIN TRANSACTION;
UPDATE Fine
SET Status = 'Unpaid'
WHERE Fine_ID = 1;
ROLLBACK TRANSACTION;
select * From Fine;

BEGIN TRANSACTION;
UPDATE Fine
SET Status = 'Paid'
WHERE Fine_ID = 1;
SAVE TRANSACTION FineSavepoint;
UPDATE Fine
SET Status = 'Unpaid'
WHERE Fine_ID = 1;
ROLLBACK TRANSACTION FineSavepoint;
COMMIT TRANSACTION;
select * From Fine;

SELECT
    Student.Student_Name,
    Book.Book_Name,
    Reservation.Reservation_ID,
```

```sql
    Reservation.Reservation_Date,
    Reservation.Status AS Reservation_Status,
    Fine.Fine_ID,
    Fine.Amount,
    Fine.Fine_Date,
    Fine.Status AS Fine_Status
FROM Student
INNER JOIN Reservation
    ON Student.Student_ID = Reservation.Student_ID
INNER JOIN Book
    ON Reservation.Book_ID = Book.Book_ID
LEFT JOIN Fine
    ON Student.Student_ID = Fine.Student_ID
    AND Book.Book_ID = Fine.Book_ID;
    SELECT
    Student.Student_ID,
    Student.Student_Name,
    COUNT(DISTINCT Reservation.Reservation_ID) AS Total_Reservations,
    COUNT(DISTINCT Fine.Fine_ID) AS Total_Fines,
    ISNULL(SUM(DISTINCT Fine.Amount), 0) AS Total_Fine_Amount
FROM Student
LEFT JOIN Reservation
    ON Student.Student_ID = Reservation.Student_ID
LEFT JOIN Fine
    ON Student.Student_ID = Fine.Student_ID
GROUP BY
    Student.Student_ID,
    Student.Student_Name;


    SELECT
    Student.Student_ID,
    Student.Student_Name,
    COUNT(Reservation.Reservation_ID) AS Total_Reservations
FROM Student
INNER JOIN Reservation
    ON Student.Student_ID = Reservation.Student_ID
GROUP BY
    Student.Student_ID,
    Student.Student_Name
HAVING COUNT(Reservation.Reservation_ID) > 1;
```